

Potenciação

$$a^2 \cdot a^5 = a^7$$

$$a^2 \cdot a^5 \cdot a^3 \cdot a^{-2} = a^{2+5+3-2} = a^8$$

$$b^2 b^{-4} \cdot b^3 \cdot b^{-2} = b^{2-4+3-2} = b^{-1} = \frac{1}{b}$$

$$C^{-3} \rightarrow \frac{1}{C^3}$$

$$C^{-n} \rightarrow \frac{1}{C^n}$$

$$a^3 : a^2 = \frac{a^3}{a^2} = a^{3-2} = a^1$$

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$$a^3 \cdot a^2 : a^2 \cdot a^3$$

$$a^5 : a^2 \cdot a^3$$

$$a^3 \cdot a^3 = a^6$$

Ejemplo :

$$2^3 \cdot 2^5 + 3^2 \cdot 3^6 : 3^2$$

$$2^8 + 3^8 : 3^2$$

$$2^8 + 3^{8-2}$$

$$2^8 + 3^6$$

$$256 + 729 = 985 //$$

$$3^3 \cdot 3^4 + 2^2 \cdot 2^6 : 5^2$$

$$3^7 + 2^8 : 5^2$$

$$2187 + 256 : 25$$

$$2187 + 10,24 = \boxed{2197,24}$$

Propiedad Distributiva

$$\boxed{(a^m)^n = a^{m \cdot n}}$$

$$(3^2)^5 = 3^{2 \cdot 5} = 3^{10}$$

$$(3^6)^{-2} = 3^{6 \cdot (-2)} = 3^{-12} = \frac{1}{3^{12}} = \frac{1}{531441} \approx 0.0000018$$

Fracciones

$$\left(\frac{a}{b}\right)^3 = \frac{a^3}{b^3}$$

$$\left[\left(\frac{a}{b}\right)^m\right]^n = \left(\frac{a}{b}\right)^{m \cdot n} = \frac{a^{m \cdot n}}{b^{m \cdot n}}$$

$$\left(\frac{2}{3}\right)^3 = \frac{2^3}{3^3}$$

$$\left[\left(\frac{2}{3}\right)^3\right]^6 = \left(\frac{2}{3}\right)^{3 \cdot 6} = \left(\frac{2}{3}\right)^{18} = \frac{2^{18}}{3^{18}} = \frac{\boxed{}}{\boxed{}}$$

$$\left[\left(\frac{1}{8} \right)^{-1} \right]^{-2} = \left(\frac{1}{8} \right)^2 = \frac{1^2}{8^2} = \boxed{\frac{1}{64}}$$

Propiedad Distributiva

$$(a \cdot b)^n = a^n \cdot b^n$$

$$(3 \cdot 6)^4 = 3^4 \cdot 6^4 = 3 \cdot 3 \cdot 3 \cdot 3 \cdot 6 \cdot 6 \cdot 6 \cdot 6 = \boxed{}$$

Ejemplos:

$$(-2a)^4 = (-2)^4 \cdot a^4 = 16a^4$$

$$(-2)^4 = (-2)(-2)(-2)(-2) = (4)(4)$$

$$\left(-\frac{3}{5} b^4 \right)^3 = \left(-\frac{3}{5} \right)^3 \cdot (b^4)^3 = \frac{(-3)^3}{5^3} \cdot b^{12} = \boxed{\frac{-27}{125} \cdot b^{12}} //$$

$$(-3)^3 = (-3)(-3)(-3) = -27$$

$$5^3 = 5 \cdot 5 \cdot 5 = 125$$

Ejemplos de Aplicación

$$\begin{aligned} & \frac{(a^3)^4 \cdot (a \cdot b^2)^3 \cdot a^{-2}}{(a \cdot b)^9} \\ & \frac{a^{12} \cdot (a \cdot b^2)^3 \cdot a^{-2}}{(a \cdot b)^9} \\ & \frac{a^{12} \cdot a^3 \cdot b^6 \cdot a^{-2}}{(a \cdot b)^9} \\ & \frac{a^{12} \cdot a^3 \cdot b^6 \cdot a^{-2}}{a^9 \cdot b^9} \\ & \frac{a^{13} \cdot b^6}{a^9 \cdot b^9} \\ & a^4 \cdot b^{-3} \\ & a^4 \cdot \frac{1}{b^3} = \boxed{\frac{a^4}{b^3}} / \end{aligned}$$

$$(a \cdot b^2)^3 = a^3 \cdot (b^2)^3 = a^3 \cdot b^6$$

$$(a \cdot b)^9 = a^9 \cdot b^9$$

$$a^{12} \cdot a^3 \cdot a^{-2}$$

$$a^{12+3-2} = a^{13}$$

$$\frac{a^{13}}{a^9} = a^{13-9} = a^4$$

$$\frac{b^6}{b^9} = b^{6-9} = b^{-3} = \frac{1}{b^3}$$

Ejemplo II

$$\frac{a^{12} \cdot a^3 \cdot b^6 \cdot a^{-2}}{a^9 \cdot b^9} = \frac{a^3 \cdot b^6}{a^9 \cdot b^9} = \frac{a^4}{b^3}$$

Ejemplo III

$$a^{12} \cdot a^3 \cdot b^6 \cdot a^{-2} \cdot a^{-9} \cdot b^{-9}$$

$$a^{12} \cdot a^3 \cdot a^{-2} \cdot a^{-9} = a^4$$

$$b^6 \cdot b^{-9} = b^{-3}$$

$$a^4 \cdot b^{-3} = \frac{a^4}{b^3}$$

Ejemplo inventado

$$(m^3)^2 \cdot (n^2 \cdot m)^3 \cdot [k^3 \cdot (m)^3]^6$$

$$\frac{[m \cdot n^2]^3 \cdot k^2}{m^6 \cdot (n^2)^3 \cdot m^3 \cdot [k^3 \cdot (m)^3]^6} = \frac{[m \cdot n^2]^3 \cdot k^2}{m^6 \cdot (n^2)^3 \cdot m^3 \cdot k^{18} \cdot (m)^{18}}$$

$$\frac{m^3 \cdot (n^2)^3 \cdot k^2}{m^6 \cdot \cancel{n^6} \cdot \cancel{m^3} \cdot k^{18} \cdot m^{18}}$$

$$\frac{\cancel{m^3} \cdot \cancel{n^6} \cdot k^2}{m^6 \cdot k^{18} \cdot m^{18}} = \frac{1}{m^6 \cdot k^{18} \cdot m^{18}}$$

$$m^6 \cdot k^{18} \cdot m^{18} =$$

$$m^{24} \cdot k^{18}$$

Radicación

$$\sqrt[n]{a^m} \rightarrow a^{m/n}$$

Ejemplos

$$\sqrt[4]{3^5} \rightarrow 3^{5/4} \approx 3$$

$$\sqrt[2]{4^6} \rightarrow 4^{6/2} = 4^3$$

$$4^{3/5} \rightarrow \sqrt[5]{4^3}$$

$$9^{2/3} \rightarrow \sqrt[3]{9^2}$$

$$16^{4/8} \rightarrow \sqrt[8]{16^4}$$

$$3^{2/7} \rightarrow \sqrt[7]{3^2}$$

Productos entre Expresiones

$$3a + 5b$$

$$-4x + 9y$$

$$a^2 \cdot (a^5) = a^7$$

$$a^2 \cdot (a^5 + a^9) = a^2 \cdot a^5 + a^2 \cdot a^9$$
$$\boxed{a^7 + a^{11}}$$

$$b^3 \cdot (b^2 + b + a^5) = b^3 \cdot b^2 + b^3 \cdot b + b^3 \cdot a^5$$

$$\boxed{b^5 + b^4 + b^3 \cdot a^5}$$

$$c^3 \cdot (b^3 + b^3 + a^5)$$

$$c^3 b^3 + c^3 b^3 + c^3 \cdot a^5$$

$$\boxed{2c^3 b^3 + c^3 \cdot a^5}$$

$$(a + b)(b^3 + b^3 + a^5)$$

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o

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